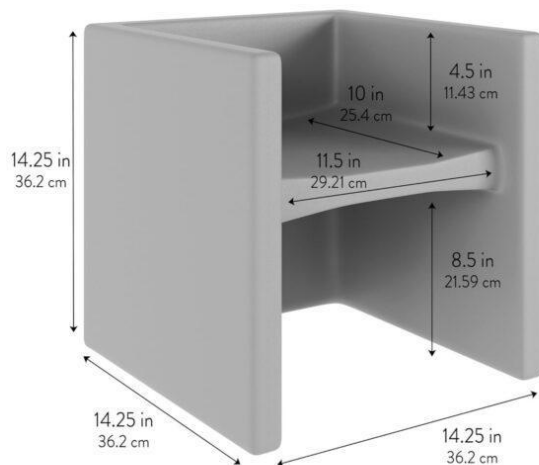


C-Tech^2 Design Project

Project Overview:

The goal of this design project is to design and build an affordable cube chair with a tray attachment. The product can be purchased online for priced anywhere between \$150 to \$225 (Links for example products in the Appendix). The cube chair is designed for children as they are developing. It can be used at home, in schools, or in physical therapy settings. It is designed with a Montessori approach, so to give a child independence, it is made to be lightweight, so the child can control where the chair is placed. Further, its wide base reduces the potential of the chair tipping over and the child injuring themselves.



User Needs	Design Inputs
Affordable	Chair w/ Tray $\leq$ \$100
Adjustable	Seat height range: 2.5" to 4.5"
Weight Bearing	Withstand a max weight of 26 lbs
Lightweight	Cannot exceed 8 lbs

User Needs & Design Inputs Notes:

- The cost will be based on the full-scale product with ideal materials
 - o If you had to use wood instead of plastic for your final design, perform your cost analysis on the plastic
- Your product MUST be scaled down by half due to time and material limitations, so plan your mechanisms accordingly

Additional Design Information from Johanna (an occupational therapist that uses this product)

- What features are essential for the chair's function?
 - o It is important that the chair has a detachable tray and that both the chair and tray are generally light and easy to maneuver. Also, the fact that the chair and tray assist in proper ergonomics is highly important (feet flat on the ground; 90/90/90 at ankles, knees, hips).
- What does a typical session look like using this chair?
 - o A typical pediatric occupational therapy session would incorporate some moving, such as completing an obstacle course, playing on a swing or a rock wall, and some seated activities such as worksheets, puzzles, sensory bin exploration, etc... We often transition from movement activities to sitting, back to movement, and then sitting throughout at 45-minute session. I would encourage your students to find some YouTube videos of a typical pediatric occupational therapy session if they need assistance visualizing what kind of kids we see.

Links to available products

- Chair with tray attachment ([link](#))
- Chair only ([link](#))
- Tray attachments ([link 1](#), [link 2](#))